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Sajad Abdulkarem Alsalem 2240007190

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Lab 6 — Filtering in the Frequency Domain

Applying edge-detection filters in the frequency domain using the Cameraman image.

Laplacian example: We resized the image to 256×256, converted it to the frequency domain using FFT, and multiplied it by a Laplacian filter. Inverse FFT returned the result to the spatial domain, highlighting edges and rapid brightness changes.

Sobel task: We padded and aligned the Sobel X and Y kernels, then transformed them using FFT. We multiplied each filter by the image's FFT and applied inverse FFT. Sobel X highlighted vertical edges, while Sobel Y highlighted horizontal edges. Combining both responses produced the overall edge magnitude.

In the End. Frequency-domain filtering applies filters through multiplication. The Laplacian and Sobel filters emphasized edges, while smooth regions produced weaker responses.

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